



**RAS P.E.**  
CONSULTING ENGINEERS  
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Project  
537-561 Johnson Avenue Project

Section  
Prop. Central Column Pilecap (310k Service)

Calc. by  
HSK

Date  
3/22/2017

Chk'd by

Date

Job Ref.  
170102

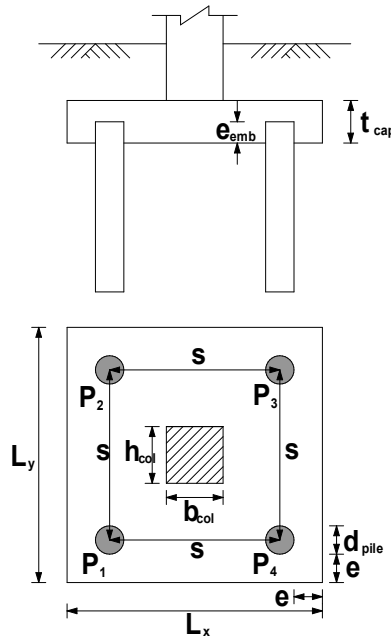
Sheet no./rev.  
1

App'd by  
Hazim K.

Date  
3/22/2017

### PILE CAP DESIGN (ACI318-11)

TEDDS calculation version 1.1.06



### Summary

#### Pile details

|                         |                       |                           |                      |
|-------------------------|-----------------------|---------------------------|----------------------|
| Number of piles         | $N = 4$               | Pile diameter             | $d_{pile} = 10.0$ in |
| Pile capacity (service) | $P_{cap} = 50.0$ kips |                           |                      |
| Pile spacing provided   | $s = 36.0$ in         | Min pile spacing required | $s_{min} = 36.0$ in  |

**PASS - Provided spacing is not less than minimum required**

|               |               |                            |                     |
|---------------|---------------|----------------------------|---------------------|
| Edge distance | $e = 15.0$ in | Min edge distance required | $e_{min} = 10.0$ in |
|---------------|---------------|----------------------------|---------------------|

**PASS - Provided edge distance is not less than minimum required**

#### Pile cap details

|                             |                    |                             |                       |
|-----------------------------|--------------------|-----------------------------|-----------------------|
| Concrete cover              | $c_c = 4.5$ in     | Overall pile cap thickness  | $t_{cap} = 44.0$ in   |
| Length of pile cap (X axis) | $L_x = 6.3$ ft     | Length of pile cap (Y axis) | $L_y = 6.3$ ft        |
| Pile embedment distance     | $e_{emb} = 8.0$ in | Min embedment required      | $e_{c\_min} = 6.0$ in |

**PASS - Provided embedment distance is not less than minimum required**

#### Column details

|                       |                       |                       |                       |
|-----------------------|-----------------------|-----------------------|-----------------------|
| Column width (X axis) | $b_{col} = 16.000$ in | Column width (Y axis) | $h_{col} = 16.000$ in |
|-----------------------|-----------------------|-----------------------|-----------------------|

#### Material properties

|                         |                                   |                   |                    |
|-------------------------|-----------------------------------|-------------------|--------------------|
| Yield strength of steel | $f_y = 60$ ksi                    | Concrete strength | $f'_c = 4.500$ ksi |
| Concrete density        | $\rho_c = 150$ lb/ft <sup>3</sup> |                   |                    |

#### Reinforcement details

|                               |                       |                                 |                       |
|-------------------------------|-----------------------|---------------------------------|-----------------------|
| Dia no. of steel bar (X axis) | $D_{x\_bar\_num} = 9$ | Spacing of steel bars ( X axis) | $s_{x\_bar} = 6.6$ in |
| Dia no. of steel bar (Y axis) | $D_{y\_bar\_num} = 9$ | Spacing of steel bars ( Y axis) | $s_{y\_bar} = 6.6$ in |

#### Loading details

|                        |                      |                        |                       |
|------------------------|----------------------|------------------------|-----------------------|
| Column axial dead load | $P_{dl} = 84.5$ kips | Column axial live load | $P_{ll} = 130.8$ kips |
|------------------------|----------------------|------------------------|-----------------------|



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| Project<br>537-561 Johnson Avenue Project              |                   |          |      | Job Ref.<br>170102   |                   |
| Section<br>Prop. Central Column Pilecap (310k Service) |                   |          |      | Sheet no./rev.<br>2  |                   |
| Calc. by<br>HSK                                        | Date<br>3/22/2017 | Chk'd by | Date | App'd by<br>Hazim K. | Date<br>3/22/2017 |

Unfactored DL mmt (X axis)  $M_{dl\_x} = 0.0$  kip\_ft

Unfactored DL mmt (Y axis)  $M_{dl\_y} = 0.0$  kip\_ft

Surcharge pressure  $p_{sur} = 0.1$  ksf

**Pile capacity check**

The service load per pile is  $P_{n\_pile} = 60.7$  kips

Unfactored LL mmt (X axis)  $M_{ll\_x} = 0.0$  kip\_ft

Unfactored LL mmt (Y axis)  $M_{ll\_y} = 0.0$  kip\_ft

**FAIL - Pile capacity is not sufficient**

**FAIL - Service load on pile is greater than pile capacity**